

SIYANG LI

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Education

Huazhong University of Science and Technology, China Ph.D. in Artificial Intelligence, School of Artificial Intelligence and Automation	2022 – 2026
Boston University, USA Master in Artificial Intelligence, College of Arts and Sciences	2020 – 2021
New York University, USA Bachelor in Mathematics and Computer Science, Courant Institute	2014 – 2018

Research Experience

Transfer Learning for Calibration-Efficient Brain-Computer Interfaces Advisor: Prof. Dongrui Wu (IEEE Fellow; EiC, IEEE Trans. Fuzzy Systems) Huazhong University of Science and Technology, China Brain-computer interface (BCI) is an interdisciplinary field covering biomedical engineering, computer science, etc. EEG -based non-invasive BCIs face challenges of individual variability and signal non-stationarity, making per-use calibration necessary yet inconvenient. My research develops transfer learning methods that align auxiliary data from other subjects, devices, or modalities with the target user, thereby reducing reliance on target-specific data. This enables fast and accurate BCI decoding with improved robustness and stronger privacy protection.	2021 – 2026
Language Modeling for Low-Resource Machine Translation Advisor: Prof. Derry Wijaya Boston University, USA Large language models (LLMs) have transformed natural language processing, but internet text corpora are dominated by high-resource languages such as English and Mandarin. My research developed transfer learning approaches to adapt pre-trained LLMs on mainstream languages to low-resource languages, enabling efficient transfer with minimal aligned corpora. This enables improved machine translation quality for data-scarce languages and scenarios with limited data.	2020 – 2021

Research Interests

My research interests include **machine learning** and its applications. Technically, I study **transfer learning**, **ensemble learning**, and **multimodal learning** methods and theories. I am also familiar with modern **large language models and vision-language models (LLMs/VLMs)**. For applications, I am particularly interested in brain decoding for **brain-computer interfaces (BCIs)**. In the long term, I aim to build **multimodal human-computer interaction** systems for future generations.

Publications

Machine Learning & AI

- [1] **S. Li†**, C. Liu†, D. Wu*, Z. Zeng*, and L. Ding*, "StackingNet: Collective Inference Across Independent AI Foundation Models," **Advanced Science**, e76488, 2026. **Paper**. **Code**.
- [2] **S. Li**, Z. Wang, C. Liu, and D. Wu*, "Black-Box Test-Time Ensemble," IEEE Computational Intelligence Magazine, vol. 21, no. 1, pp. 57-68, 2026. **Code**.
- [3] Z. Cui, **S. Li**, S. Kan, Y. Wang, F. Luo, D. Wu*, "A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces," IEEE Transactions on Emerging Topics in Computational Intelligence, early access, 2026.
- [4] H. Wang, H. Zhang, **S. Li**, D. Wu*, "Gated Parametric Neuron for Spike-based Audio Recognition," Neurocomputing, vol. 609, pp. 128477, 2024. **Code**.
- [5] G. Kuwanto†, A. F. Akyürek†, I. C. Tourni†, **S. Li†**, A. Jones, and D. Wijaya*, "Low-Resource Machine Translation Training Curriculum Fit for Low-Resource Languages," in Proc. Pacific Rim International Conference on Artificial Intelligence, Jakarta, Indonesia, Nov. 2023. [co-first author]
- [6] H. Wu, **S. Li**, and D. Wu*, "TMMM: Transformer in Multimodal Sentiment Analysis under Missing Modalities," Proc. International Joint Conference on Neural Networks, Yokohama, Japan, Jun. 2024.
- [7] **S. Li**, Y. Xu, H. Wu, D. Wu*, Y. Yin, J. Cao, and J. Ding, "Facial Expression Recognition In-the-Wild with Deep Pre-trained Models," in Proc. European Conf. on Computer Vision Workshop, Tel Aviv, Israel, Oct. 2022. **Code**.

- [1] **S. Li**, Z. Wang, H. Luo, L. Ding*, and D. Wu*, "T-TIME: Test-Time Information Maximization Ensemble for Plug-and-Play BCIs," *IEEE Transactions on Biomedical Engineering*, vol. 71, no. 2, pp. 423–432, 2024. [Code](#).
- [2] **S. Li**, Z. Wang, S. Zheng, and D. Wu*, "Heterogeneous Supervised Domain Adaptation (HSDA) for Cross-Device EEG Classification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, R3, under review. [Code](#).
- [3] **S. Li**, H. Wu, L. Ding, and D. Wu*, "Meta-Learning for Fast and Privacy-Preserving Source Knowledge Transfer of EEG-Based BCIs," *IEEE Computational Intelligence Magazine*, vol. 17, no. 4, pp. 16–26, 2022. [Code](#).
- [4] **S. Li†**, J. Ouyang†, Z. Cui†, Z. Wang, T. Jia, F. Wan, and D. Wu*, "Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces," *IEEE Journal of Biomedical and Health Informatics*, R1, under review. [Code](#).
- [5] **S. Li**, H. Wang, X. Chen, and D. Wu*, "Multimodal Brain-Computer Interfaces: AI-powered Decoding Methodologies," *arXiv preprint*, 2025. [arXiv](#).
- [6] Z. Wang†, **S. Li†**, J. Luo, J. Liu, D. Wu*, "Channel Reflection: Knowledge-Driven Data Augmentation for EEG-based Brain-Computer Interfaces," *Neural Networks*, vol. 176, pp. 106351, 2024. [co-first author] [Code](#).
- [7] H. Wu, **S. Li**, and D. Wu*, "Motor Imagery Classification for Asynchronous EEG-Based Brain-Computer Interfaces," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 527–536, 2024. [Code](#).
- [8] X. Chen, **S. Li**, D. Wu*, "AFPM: Alignment-based Frame Patch Modeling for Cross-Dataset EEG Decoding," *Science China Information Sciences*, vol. 69, no. 6, 162403, 2026. [Code](#).
- [9] D. Liu, **S. Li**, Z. Wang, W. Li, D. Wu*, "Spatial Distillation based Distribution Alignment (SDDA) for Cross-Headset EEG Classification," *IEEE Transactions on Biomedical Engineering*, early access, 2025. [Code](#).
- [10] Z. Wang, **S. Li**, X. Chen, and D. Wu*, "Time-Frequency Transform based EEG Data Augmentation for Brain-Computer Interfaces," *Knowledge-Based Systems*, vol. 311, pp. 113074, 2025. [Code](#).
- [11] Z. Wang, **S. Li**, X. Chen, and D. Wu*, "MVCNet: Multi-View Contrastive Network for Motor Imagery Classification," *Knowledge-Based Systems*, vol. 328, pp. 114205, 2025. [Code](#).
- [12] Z. Wang, H. Wang, T. Jia, X. He, **S. Li**, D. Wu*, "DBConformer: Dual-Branch Convolutional Transformer for EEG Decoding," *IEEE Journal of Biomedical and Health Informatics*, vol. 30, no. 5, pp. 4134–4147, 2026. [Code](#).
- [13] X. Chen, J. An, H. Wu, **S. Li**, B. Liu, and D. Wu*, "Front-End Replication Dynamic Window (FRDW) for Online Motor Imagery Classification," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 3906–3914, 2023. [Code](#).
- [14] C. Liu, **S. Li**, L. Tan, and D. Wu*, "Temporal Out-of-Distribution Detection for Asynchronous Motor Imagery Brain-Computer Interfaces," *arXiv preprint*, 2026. [arXiv](#). [Code](#).

Clinical & Trustworthy Brain Decoding

- [1] Z. Wang, **S. Li**, D. Wu*, "Canine EEG Helps Human: Cross-Species and Cross-Modality Epileptic Seizure Detection via Multi-Space Alignment," *National Science Review*, vol. 12, no. 6, nwafo86, 2025. [Code](#).
- [2] **S. Li†**, Z. Wang†, X. Gui†, X. Chen, Z. Wang, Y. Wen, and D. Wu*, "RAICL: Retrieval-Augmented In-Context Learning for Vision-Language-Model Based EEG Seizure Detection," *IEEE Transactions on Systems, Man and Cybernetics: Systems*, under review, preprint available: [arXiv](#). [Code](#).
- [3] Z. Wang, W. Zhang, **S. Li**, X. Chen, and D. Wu*, "Unsupervised Domain Adaptation for Cross-Patient Seizure Classification," *Journal of Neural Engineering*, vol. 20, no. 6, pp. 066002, 2023. [Code](#).
- [4] H. Wang, Z. Jia, Y. Shen, Z. Wang, **S. Li**, Z. Tang, K. Shu, F. Hu, D. Wu*, "SACM: SEEG-Audio Contrastive Matching for Chinese Speech Decoding," *IEEE Transactions on Biomedical Engineering*, early access, 2026. [Code](#).
- [5] Y. Tu, **S. Li**, X. Chen, D. Wu*, "BrainprintNet: A Multiscale Cross-Band Fusion Network for EEG-based Brainprint Recognition," *IEEE Transactions on Information Forensics and Security*, vol. 21, pp. 2757–2768, 2026. [Code](#).
- [6] X. Chen, **S. Li**, Y. Tu, Z. Wang, D. Wu*, "User-wise Perturbations for User Identity Protection in EEG-Based BCIs," *Journal of Neural Engineering*, vol. 22, no. 1, pp. 016040, 2025. [Code](#).
- [7] T. Jia, L. Meng, **S. Li**, D. Wu*, "Federated Motor Imagery Classification for Privacy-Preserving Brain-Computer Interfaces," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 3442–3451, 2024. [Code](#).
- [8] X. Jiang, L. Meng, **S. Li**, and D. Wu*, "Active Poisoning: Efficient Backdoor Attacks on Transfer Learning-based Brain-Computer Interfaces," *Science China Information Sciences*, vol. 66, no. 8, 182402, 2023.

Awards

- Hua-Nao (China-Brain) Award, Top-10 Highlights in China's Brain-Computer Interface, *Transfer Learning for Accurate EEG Decoding*, Contributors: Dongrui Wu, Ziwei Wang, **Siyang Li**, Tianwang Jia. 2026. [News](#)
- Hua-Nao (China-Brain) Award, Top-10 Highlights in China's Brain-Computer Interface, *Knowledge-Data Fusion for Accurate and Efficient Brain-Computer Interface Decoding*, Contributors: Dongrui Wu, Xue Jiang, Ziwei Wang, Ruimin Peng, Lubin Meng, **Siyang Li**. 2024. [News](#).
- National Scholarship (Doctoral Students), 2024.
- Youth Paper Defense Award, World Robot Contest - Brain-Computer Interface Track, 2023.

Funding

- 20,000 CNY. Grant from the Henan Key Laboratory of Brain Science and Brain-Computer Interface Technology Open Fund (HNBBL230204), *Research on Transfer Learning Algorithms for Calibration-Free Brain-Computer Interfaces*, 2023–2025. [URL](#).

Talks

- **S. Li**, D. Wu, Tutorial: Machine Learning Pipeline for EEG-based Brain-Computer Interfaces, International Joint Conference on Neural Networks, Gold Coast, Australia, 2023.
- D. Wu, Z. Wang, **S. Li**, Tutorial: Transfer Learning for EEG-based Brain-Computer Interfaces, International Conference on Neural Information Processing, Changsha, China, 2023.

Academic Service

Reviewer for IEEE Transactions on Biomedical Engineering, IEEE Transactions on Neural Systems and Rehabilitation Engineering, IEEE Computational Intelligence Magazine, IEEE Transactions on Consumer Electronics, and related journals.

Patents

- D. Wu, **S. Li**, H. Wu. Meta-learning methods, EEG signal recognition methods, and devices for brain-computer interfaces. Chinese Invention Patent, CN114879838A, *granted*.
- D. Wu, **S. Li**, Z. Wang, S. Zheng. Construction method, classification method, and system for cross-device EEG classification models. Chinese Invention Patent, CN119719914A, *granted*.
- D. Wu, **S. Li**. Test-time adaptation methods and systems for calibration-free brain-computer interfaces. Chinese Invention Patent, CN117056837A, *granted*.
- D. Wu, **S. Li**, Z. Wang. Test-time ensemble method, medium, and device for black-box models. Chinese Invention Patent, CN118626848A, *published*.
- D. Wu, H. Wu, **S. Li**. Asynchronous motor imagery BCI recognition methods and systems. Chinese Invention Patent, CN117034105A, *granted*.
- D. Wu, X. Jiang, L. Meng, **S. Li**. Training methods and devices for machine learning models with backdoors, and electronic equipment. Chinese Invention Patent, CN114021626A, *published*.
- D. Wu, Z. Wang, **S. Li**, X. Chen. Supervised contrastive learning methods and devices based on spatio-temporal-spectral consistency. Chinese Invention Patent, CN118626901A, *published*.
- D. Wu, **S. Li**. Stroke rehabilitation methods based on the integration of motor imagery BCI and transcranial direct current stimulation. Chinese Invention Patent, CN120420602A, *published*.
- D. Wu, D. Liu, **S. Li**, Z. Wang. Construction methods and classification methods and systems for EEG models across different devices. Chinese Invention Patent, CN120067794A, *published*.